

Kidney Disease and its Impact on Oral Health

OMFS 512: Integrated Biomedical and Clinical Sciences

Rashidah T. Wiley, DDS

Mironda Frankenberger, DDS



Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

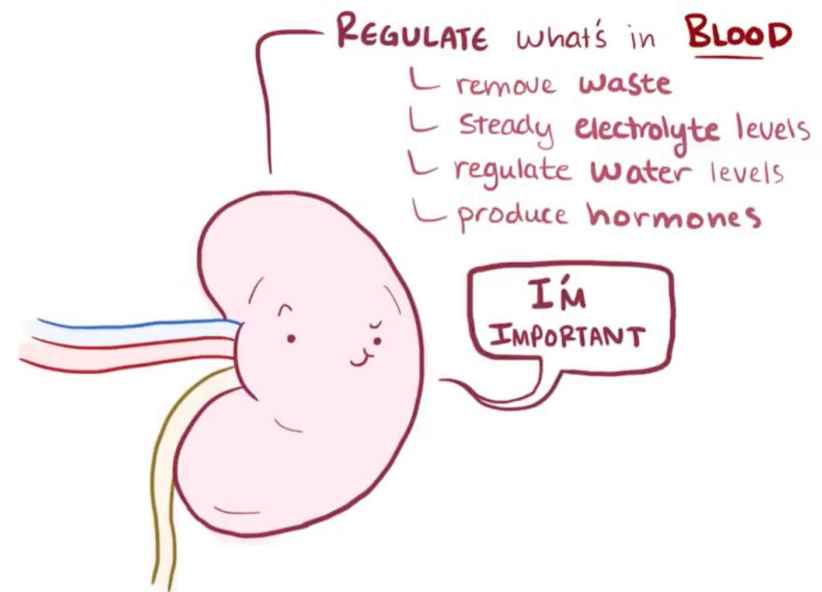
1. Review the normal function of the kidney.
2. Analyze the different forms of kidney disease.
3. Discuss the dental modifications for patients with diseases of the kidney.

Competency statements met

#1.2,1.3, 5.1

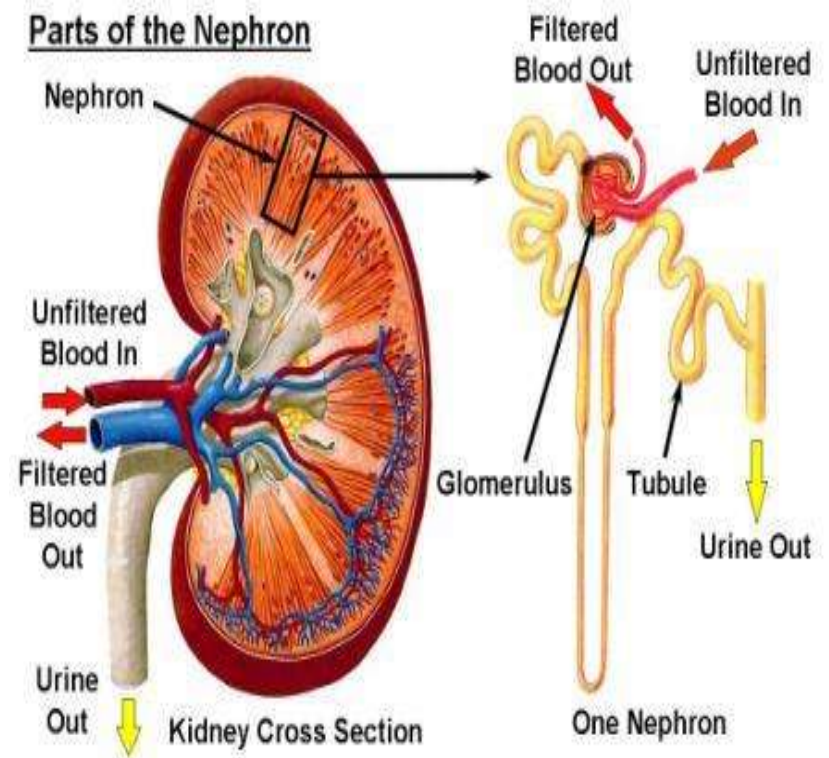
Functions of the kidney

- Filtration of waste and toxins from the body.
- Maintenance of acid-base balance .
- Regulation of fluid volume.
- Synthesis and release of hormones.
 - Erythropoietin
 - 1,25 dihydroxycholecalciferol
 - Renin



Topics of Discussion

- Nephrolithiasis (Kidney stones)
- Acute Kidney Injury
- Polycystic Kidney Disease
- Glomerular disorders
 - Nephrotic vs Nephritic syndrome
 - Chronic Kidney Disease (Chronic Renal Failure)
- Renal Cancer
- Kidney Transplant
- Dental Considerations



unckidneycenter.org/kidneyhealthlibrary/glomerular-disease/

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Nephrolithiasis

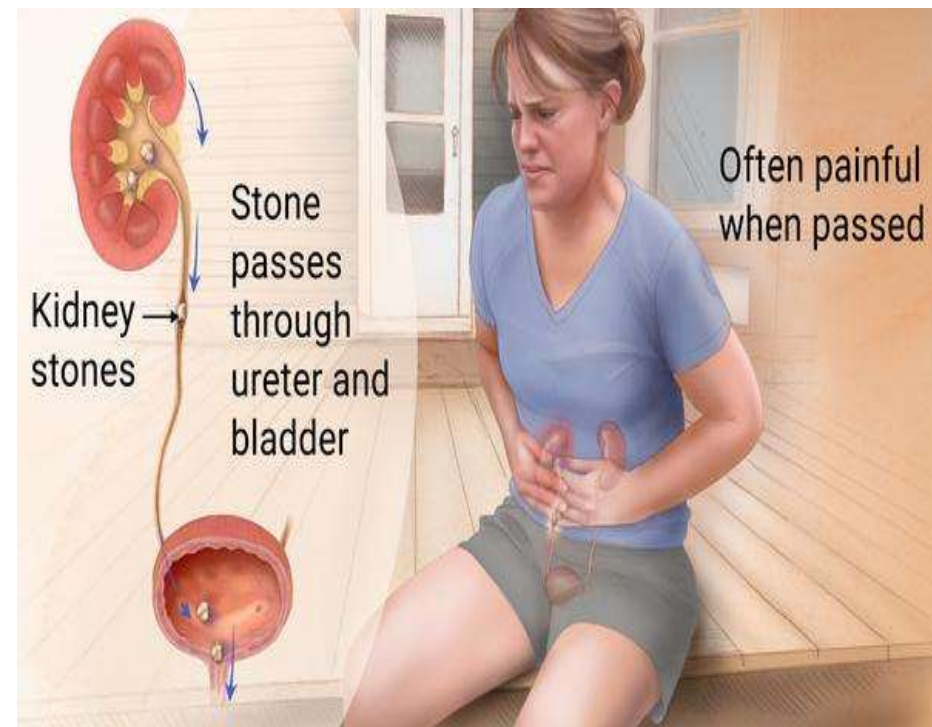
Stones forming in the pelvis of the kidney

- **Composition**

- Calcium (80% of cases)
- Uric acid
- Struvite (association with bacterial infections)
- Cystine

- **Clinical manifestations**

- **Renal colic**- when symptoms become present
 - Pain in the back, abdomen and groin regions
- **Hematuria**- blood in urine
- **Pyelonephritis**- acute or chronic infection
- **Hydronephrosis**- fluid retention in the kidneys due to blockage of the ureter leading to decreased urine output

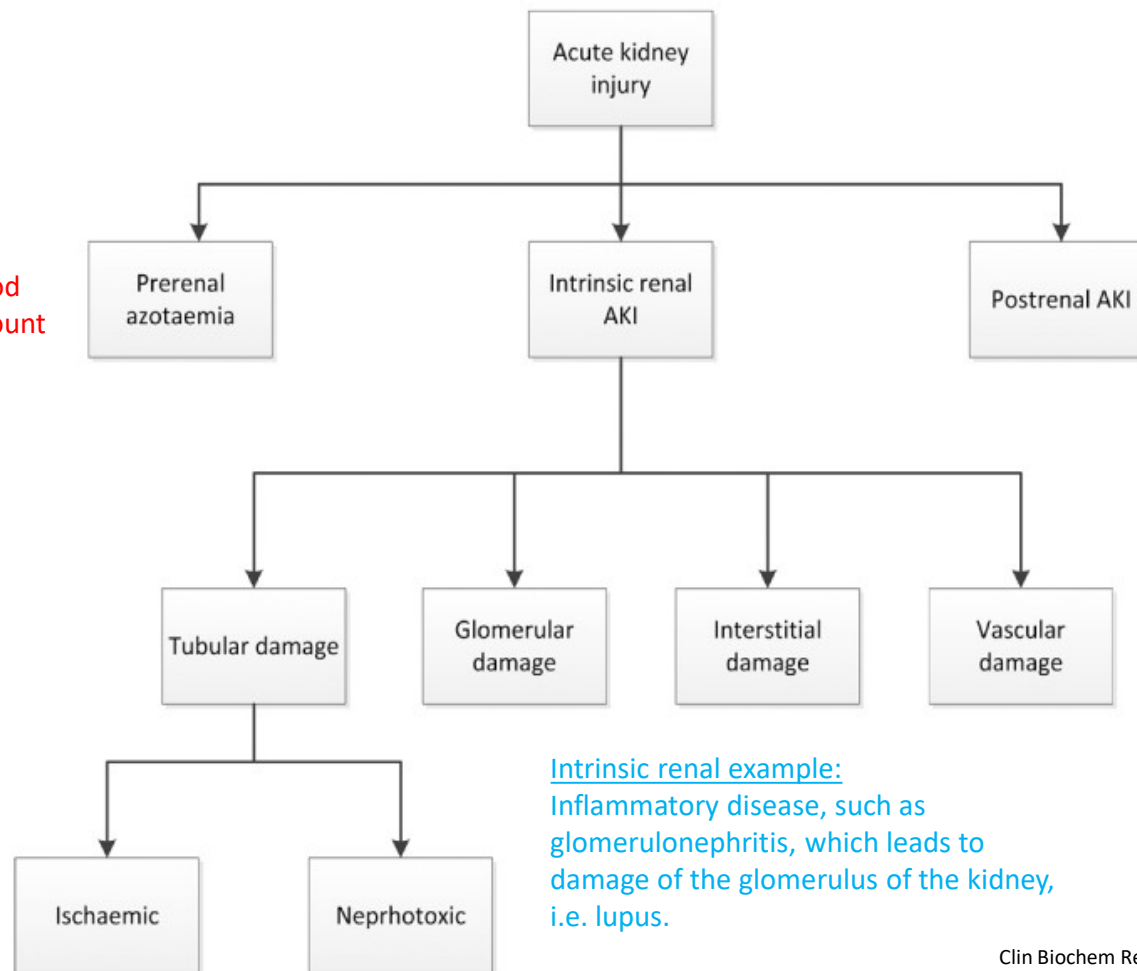


Mayo Clinic

Acute Kidney Injury

- Formally known as acute renal failure and acute tubular necrosis
- Sudden loss in the kidneys ability to function
 - An accumulation of toxic nitrogen compounds in the blood
 - Decrease in urine output
 - Also, effects electrolyte balance and maintenance of fluids
- Causes
 - Hypoperfusion of the kidney, bacterial infections, damage to the parenchyma, and obstruction of the urinary tract

Acute kidney injury



Prerenal example:
traumatic event leading to low blood volume leading to a decreased amount of blood reaching the kidney.

Postrenal example:
Obstruction
Renal stones or cancer of the bladder, prostate, or cervix.

Intrinsic renal example:
Inflammatory disease, such as glomerulonephritis, which leads to damage of the glomerulus of the kidney, i.e. lupus.

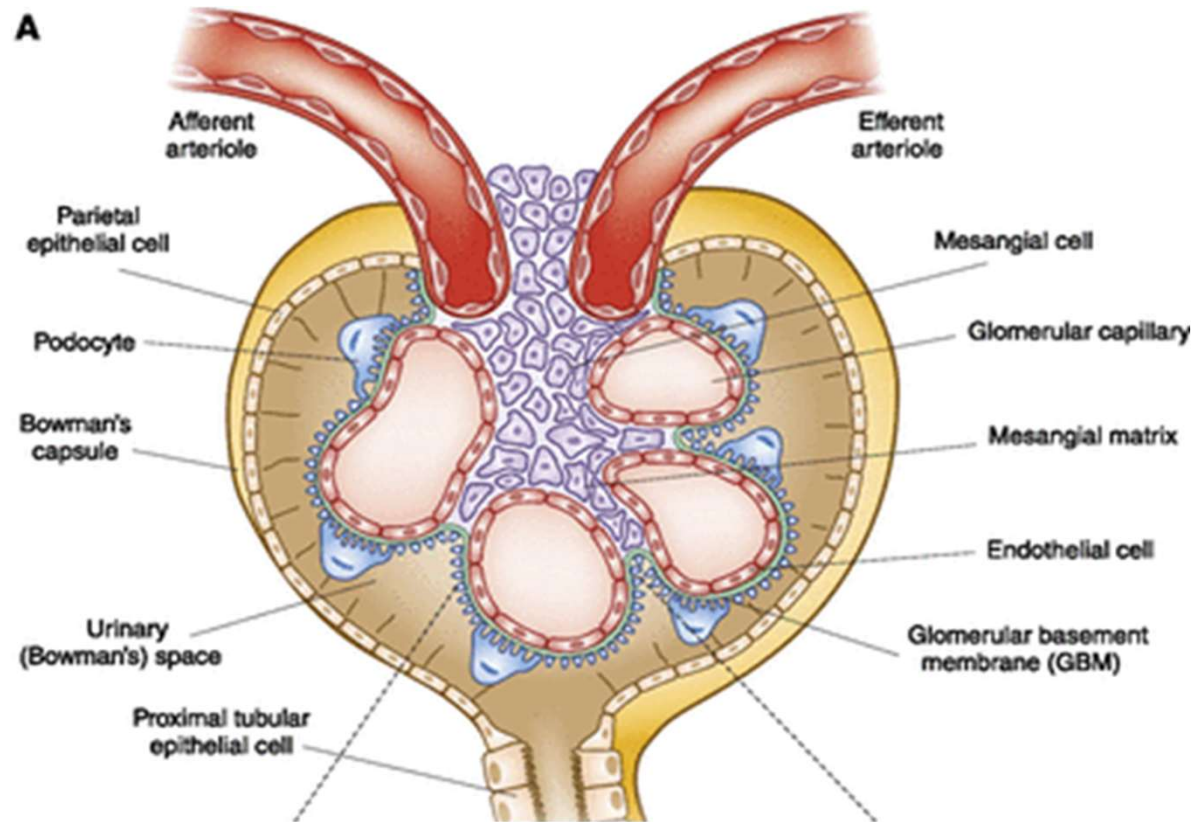
Polycystic Kidney Disease

- Autosomal dominant – Adult variant
- Autosomal recessive - childhood
- Bilateral, multiple expanding cysts that destroy the renal parenchyma
- First symptoms occur in the 4th decade
 - Severe flank pain, hematuria
- Kidneys are markedly enlarged
- End-stage renal failure by age 50

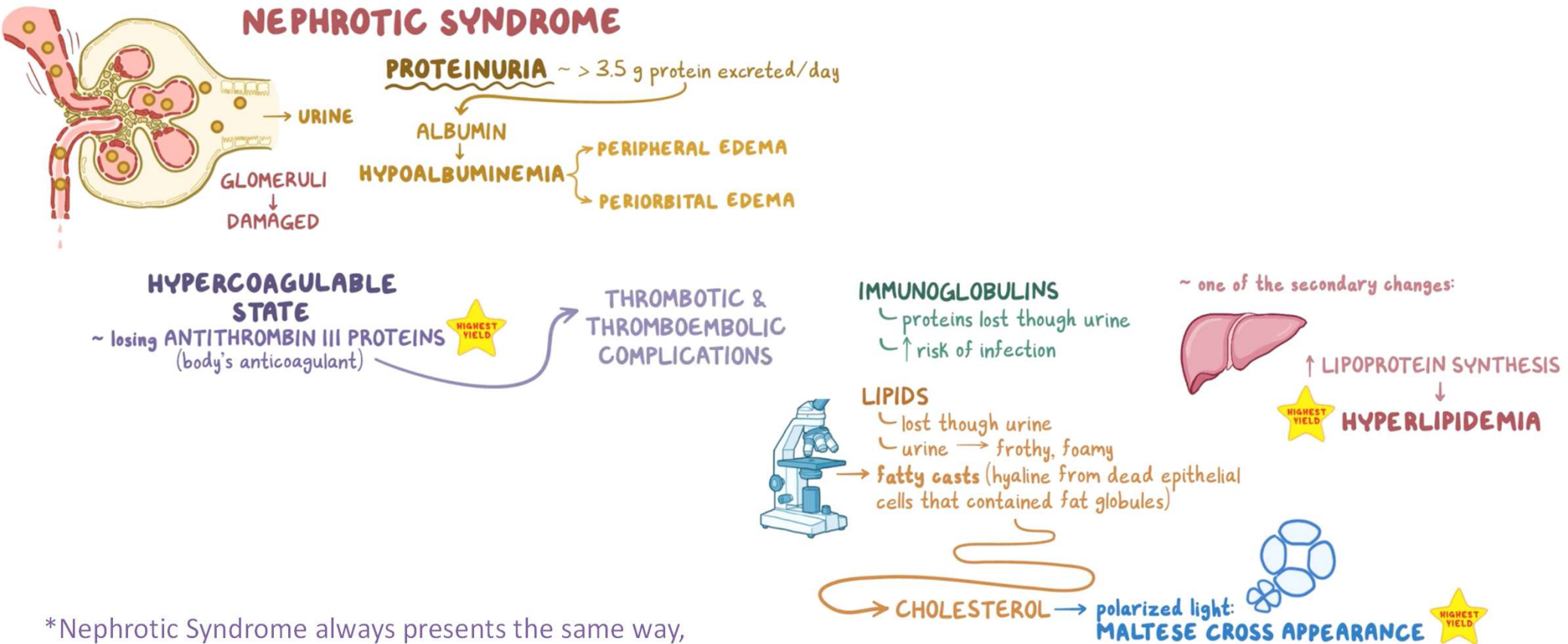


Maxeiner A, et al. Native Nephrectomy before and after Renal Transplantation in Patients with Autosomal Dominant Polycystic Kidney Disease (ADPKD). Journal of Clinical Medicine. 2019; 8(10):1622.

Glomerular Disorders of the Kidney

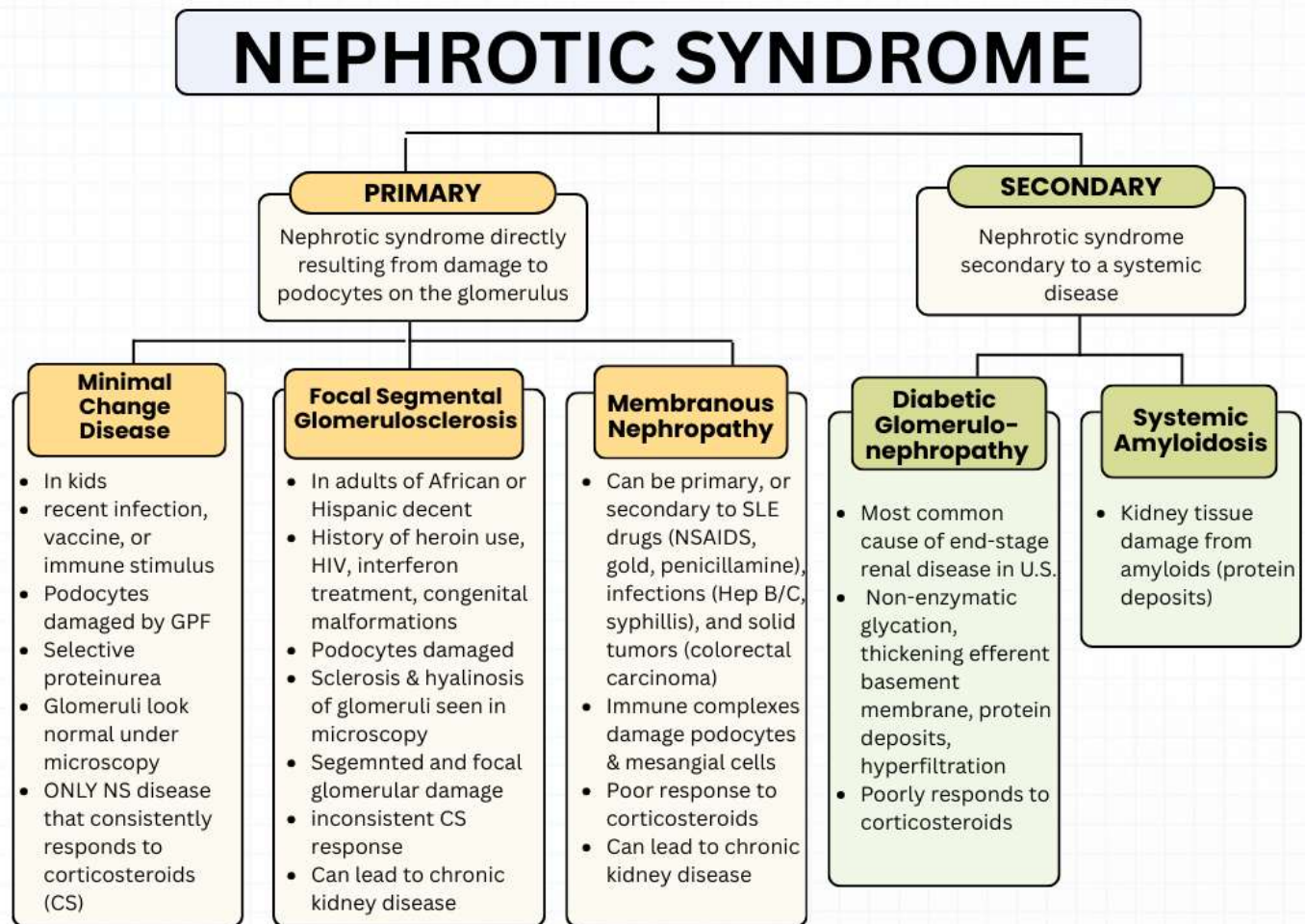


Nephrotic Syndrome- Pathophysiology



*Nephrotic Syndrome always presents the same way, regardless of the cause of the disease.

Nephrotic Syndrome Classifications



Nephritic Syndrome

NEPHRITIC SYNDROME

↳ INFLAMMATION that DAMAGES the GLOMERULAR BASEMENT MEMBRANE

↳ HEMATURIA & RBC CASTS in URINE

↳ RENAL FAILURE:

- * oliguria
- * arterial hypertension (due to Na^+ retention)
- * peripheral & periorbital edema

LAB TESTS: ↑ levels of BUN & creatinine

URINALYSIS: * hematuria

* proteinuria → **< 3.5 g/day**

* RBC casts in urine

"SUBNEPHROTIC RANGE"

* severe cases → proteinuria > 3.5 g/day

Disorders that Cause Nephritic Syndrome

TYPE III HYPERSENSITIVITY REACTION

- ~ Poststreptococcal glomerulonephritis
- ~ IgA nephropathy
- ~ diffuse proliferative glomerulonephritis

MULTIPLE MECHANISMS

- ~ MPGN
- ~ RPGN

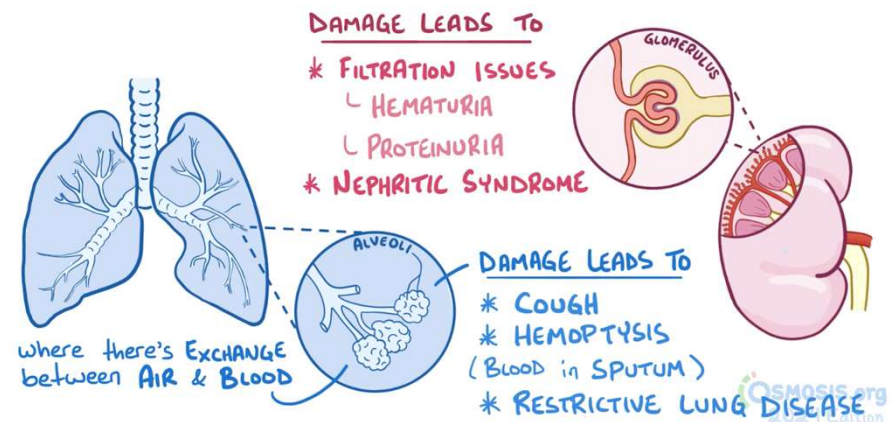
ALPORT SYNDROME

- ~ collagen synthesis

* CAREFUL HISTORY & KIDNEY BIOPSY

Important Cause of Nephritic Syndrome

- Goodpasture diseases
 - Occurs when there are IgG and complement deposits bind to the renal and pulmonary basement membranes leading to significant inflammation.
 - Individuals with Goodpasture disease can present with rapid progressive glomerular nephritis
 - Rapid loss in kidney function within a short period of time (matter of days).
 - Death may occur without intervention.
 - Aggressive treatment with corticosteroids, immunosuppression, and plasmapheresis can decrease risk of chronic renal failure



How to Differentiate Nephrotic vs Nephritic

Nephritic

Nephrotic

Components loss in urine

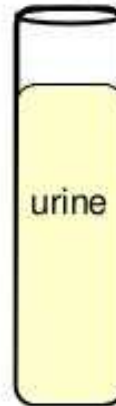
1. Protein (less than what is seen in nephrotic syndrome)
2. Red and white blood cells

The urine will have a brownish-red appearance and granular "Coca-Cola"



- Oliguria
- Hematuria
- Non selective Proteinuria.
- $GFR \downarrow$, $Cr \uparrow$, $BUN \uparrow$
- Edema (salt and water retention)
- Hypertension
- RBC & Protein casts.

- Proteinuria ("nephrotic range" $>3.5g/24h$)
- Edema (retention+Hypoalbuminemia)
- Hyperlipidemia
- Lipiduria
- Protein casts.



Components loss in urine

1. Albumin (Protein)
2. Antithrombin II (Protein)

(increased risk to thrombus formation)

3. Immunoglobulins

(increased risk of infection)

4. Lipids

The urine will have a frothy appearance

Diseases Associated with Nephrotic/Nephritic Syndromes

Nephrotic Syndrome	POC	Nephritic Syndrome
Edema - Hypo-albuminemia Proteinuria Hypercholesterolemia (not essential)	Characteristic	HTN – Mild Edema – Oliguria - Azotemia - High JVP – Dark Urine
Any Inflammation cause Proteinuria then Edema then Hypovolemia then more hypoalbuminemia and hyperlipidemia	Mechanism	2 <i>Immunological</i> disorder Causing Thickening of BM 1. Anti-Basement Membrane antibody 2. Immune Complex (ppt of BM)
High (dt: High oncotic pressure)	GFR	Low
- Minimal Change Disease - Membranous glomerulonephritis - Systemic vasculitis - Diabetic Glomerulosclerosis - Amyloidosis - Drugs: Penicillamine Captopril – Gold – Tetracycline (outdated) - Sarcoidosis - Right sided Heart failure - Constrictive pericarditis	Causes	<u>Mostly due to</u> - Acute Post-streptococcal Glomerulonephritis - Glomerulonephritis in the course of collagen disorder as SLE. <u>May be rarely due to</u> - Viral infection: Hepatitis – Epstein Barr virus – Cytomegalovirus - Parasitic infection of glomerulus: Plasmodium falciparum. - Rapidly progressive GN - Acute tubule-interstitial nephropathy.

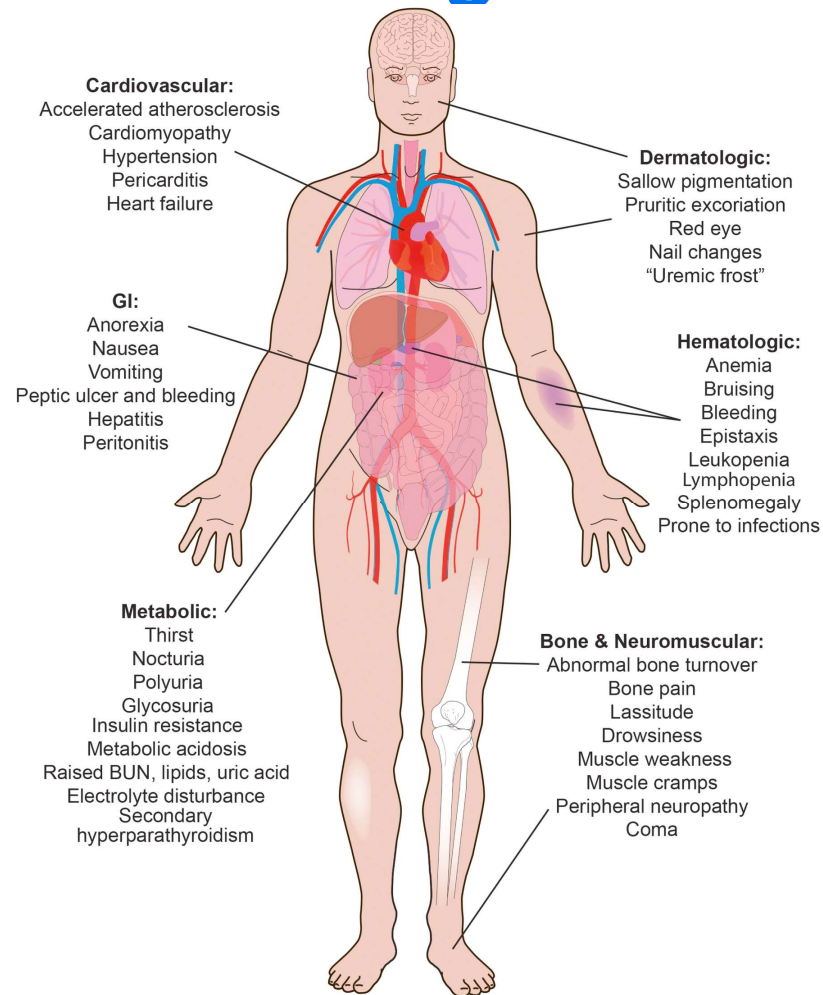
Chronic Kidney Disease (CKD)

- (Called Chronic Renal Failure at end stages of disease progression.)
- Abnormalities in the kidney structure or function resulting in a decrease in kidney function present for more than 3 months
- Risk factors: diabetes mellitus, hypertension, glomerulonephritis, polycystic kidney disease, male predilection, and race (Asian, Native American, and African decent)
 - Less common: lupus, rheumatoid arthritis, HIV, long-term NSAID use, toxins (tobacco), silica exposure, tubular interstitial nephritis, and kidney neoplasms
 - In pediatric population: congenital disease, birth defects, infection, nephrotic syndrome, trauma, and urine blockage

CKD Progression and Treatment

- Progressive disease of the kidney that results in end stage renal disease.
- Treatment involves lifestyle changes, managing underlying cause (usually a co-morbid chronic disease), dialysis, or renal transplant.

Clinical Signs of CKD/Chronic Renal Failure



Nausea and vomiting

Dry and itchy skin

Polyuria

Hematuria

Frothy urine

Puffy eyes

Swollen extremities

Muscle cramping

Bone changes

Neurologic issues

Laboratory Values for Renal Function & Failure

Table 12.2

Laboratory Values for the Assessment of Renal Function and Failure

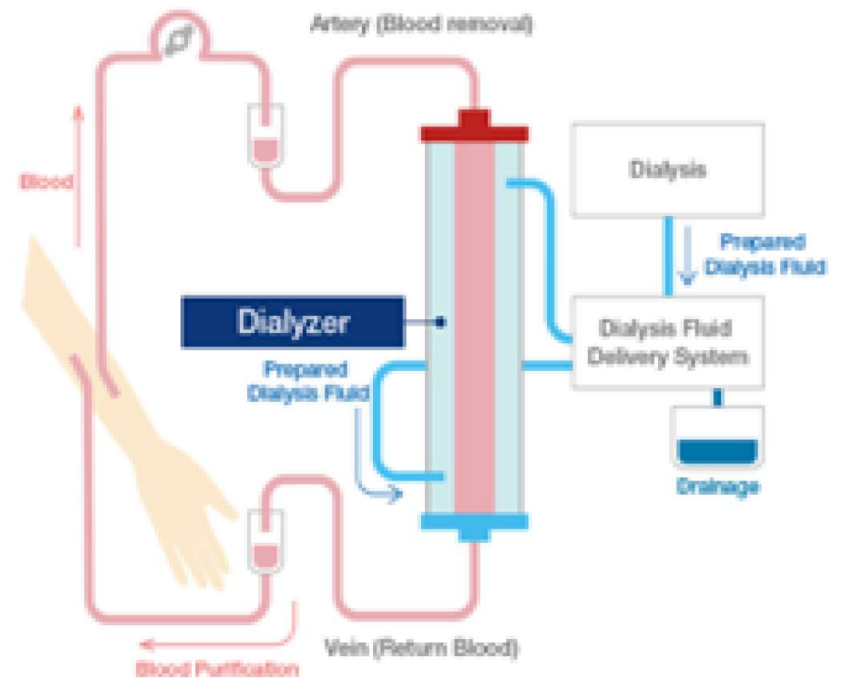
Laboratory Test	Reference Value	Indicator* of Renal Insufficiency (Stages II–IV)	Indicator of Renal Failure (Stage V)
Urine			
Albuminuria	<30mg/g	30–300mg/g	>300mg/g
Creatinine clearance (CrCl)	85–125mL/min (women) 97–140mL/min ³⁴	50–90mL/min	Moderate: 10–50mL/min; severe: <10mL/min
Glomerular filtration rate (GFR) [†]	100–150mL/min	15–89mL/min	Moderate: <15mL/min; severe: <10mL/min
Serum			
Blood urea nitrogen (BUN)	8–18mg/dL (3–6.5mmol/L)	20–30mg/dL	Moderate: 30–50mg/dL; severe: >50mg/dL
Creatinine	0.6–1.20mg/dL	2–3mg/dL	Moderate: 3–6mg/dL; severe: >6mg/dL

* Secondary indicators of renal function. Normal reference values: calcium, 8.2–11.2mg/dL; chloride, 95–103mmol/L; inorganic phosphorus, 2.7–4.5mg/dL; potassium, 3.8–5mmol/L; sodium, 136–142mmol/L; total carbon dioxide for venous blood, 22–26mmol/L; and uric acid, 2.4–7.0mg/dL.

Type of Dialysis



Peritoneal Dialysis



Dialyzer Structure

jokoh.com/en/products/dialysis-information-room

Oral Manifestations of Chronic Kidney Disease

- Gingivitis and periodontitis
- Xerostomia
- Altered taste
- Candidiasis
- Bone changes in the jaw
- Gingival hyperplasia
- Uremic stomatitis
- Oral lesions

Oyetola EO, Owotade FJ, Agbelusi GA, Fatusi OA, Sanusi AA. Oral findings in chronic kidney disease: implications for management in developing countries. BMC Oral Health. 2015 Feb 20;15:24. doi: 10.1186/s12903-015-0004-z. PMID: 25888327; PMCID: PMC4350651.

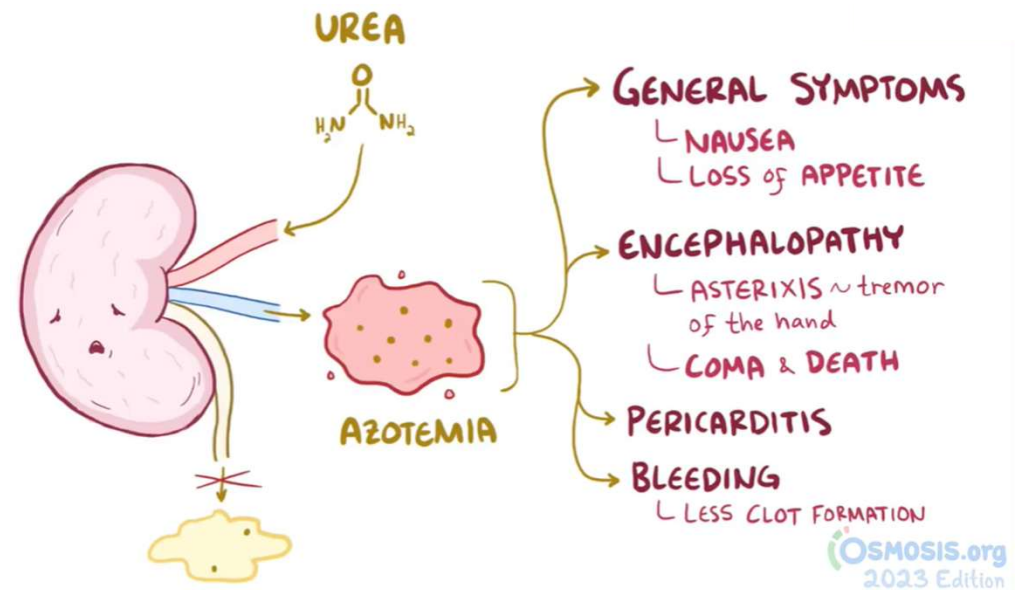
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Uremic Stomatitis

- Rare mucosal disorder associated with renal disorders
- Thick greyish-white plaques on the soft tissue of the oral cavity
- Result of uremia in CKD



Courtesy of Dr. Edmund Cataldo



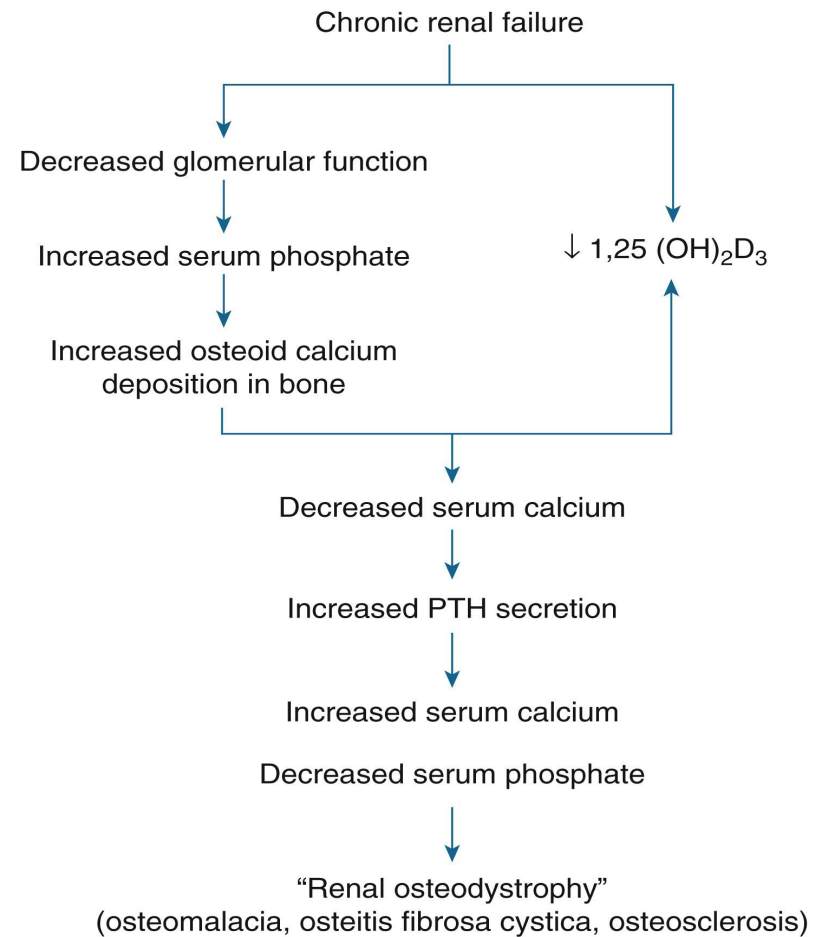
Changes in Bone

Bone remodeling can occur because of low levels of active vitamin D and serum calcium.

Active vitamin D is needed to absorb calcium in from the GI tract.

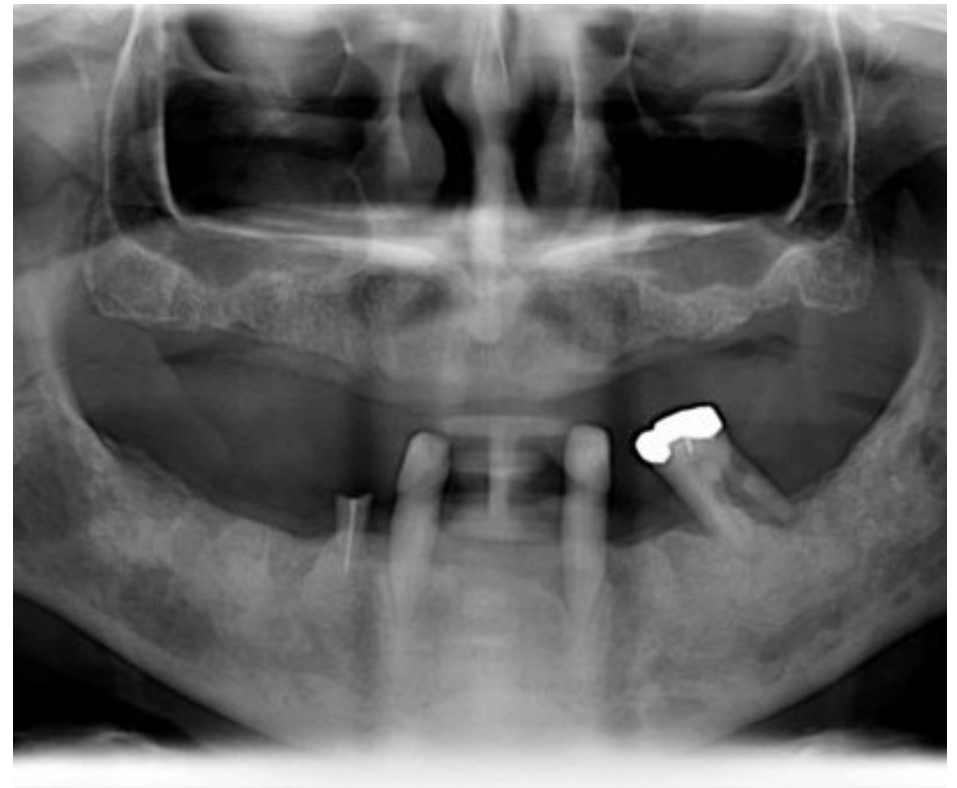
Calcium is removed from the bone when serum calcium is low. This creates bone demineralization.

Secondary Hyperparathyroidism/ Brown's Tumor of the Bone



Secondary Hyperparathyroidism

1. Loss of lamina dura
2. Bone demineralization
 - “Ground-glass” appearance
3. Expansive radiolucent lesions
 - Brown’s Tumor of Bone



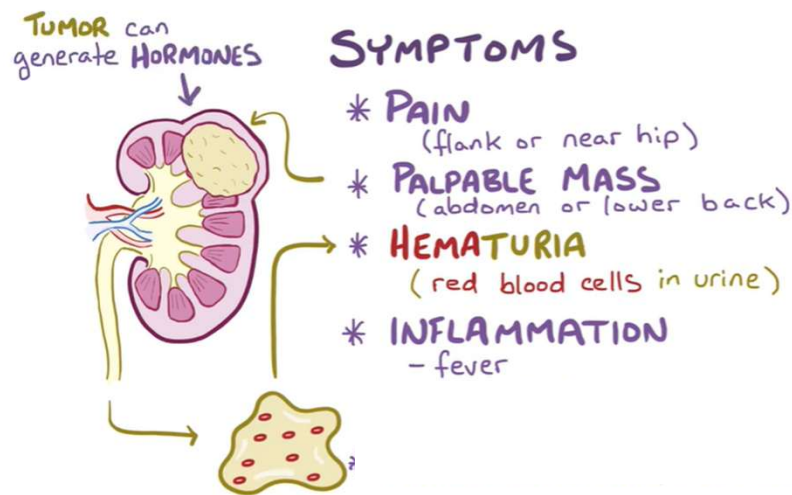
Hyperparathyroidism - Brown's Tumor of Bone



Renal Cancer

Renal Cell Carcinoma

RENAL CELL CARCINOMAS (RCC'S)



OSMOSIS.org
2023 Edition

- Adenocarcinoma of tubular epithelial cells
 - >80% of all renal malignancies
- Increased risk: older men, people who smoke, cadmium exposure, acquired polycystic disease
- Present with hematuria (50%), fever, and flank pain
- Metastasis may be first sign (lungs, bones)
 - Can also metastasize to the oral cavity

Renal Cell Carcinoma Treatment

- Resistant to radiation therapy and chemotherapy
- Kidney resection is a possibility if the tumor is localized
- Immunomodulatory agents may be effective
 - Treatment with chemokines and antibodies
- Molecular targeted therapies
 - Inhibiting tumor vascularization

Kidney Transplant

Kidney Transplant

- Surgery to replace a failing kidney/ kidney with end-stage renal disease with a healthy kidney
- Types
 - Deceased-donor kidney transplant
 - From recently deceased human
 - >2/3 of kidney transplants in the U.S.
 - Living-donor kidney transplant
 - From a living person; Only need 1 functioning kidney to live
 - ~1/3 of kidney transplants in the U.S.
 - Preemptive kidney transplant
 - Takes place prior to dialysis
 - 2.5% of kidney transplants in the U.S.

Kidney Transplant vs Dialysis

- Benefits
 - Improved quality of life
 - Lower risk of death
 - Fewer dietary restrictions
 - Lower treatment cost
- Contraindications
 - Advanced age
 - Severe heart disease
 - Active or recently treated cancer
 - Dementia or poorly controlled mental illness
 - Alcohol or drug misuse
 - Other medical contraindications

Immunosuppressant Drugs for Kidney Transplants

Table 5.3. Transplantation Immunosuppressant Medications

Medication	Action	Oral Side Effects/Dentally Relevant Toxicities
Glucocorticoids	Glucocorticoid receptor agonists, receptor-independent effects	Angioedema, moon face, oral candidiasis and oral herpes simplex virus, anemia, neutropenia, possible adrenal suppression
Cyclosporine	Calcineurin inhibitor	Gingival overgrowth
Tacrolimus	Calcineurin inhibitor	Gingival overgrowth (lower incidence than cyclosporine)
Sirolimus, everolimus	Target of rapamycin inhibitors	Mouth ulcers, delayed wound healing, thrombocytopenia
Mycophenolate mofetil	Purine synthesis inhibitor	Neutropenia, mild anemia
Azathioprine	Antimetabolite	Leukopenia, bone marrow depression
Daclizumab, basiliximab	Anti-CD25 antibody	Rare mouth ulcers, oral candidiasis, gingival overgrowth, thrombocytopenia
Horse or rabbit antithymocyte globulin	Polyclonal antithymocyte globulin	Thrombocytopenia, leukopenia

Adapted from Hallorhan PF. N Engl J Med 2004;351(26): 2715-29.

Dental Considerations

Dental Considerations for Renal Disease

- Dentist should perform patient screening
 - Stage of renal disease and stage
 - Does the patient have any other comorbidities associated with renal disease?
 - Treatment the patient is receiving
 - Does the patient require prophylaxis antibiotics before treatment?
 - Are there any drugs which should be avoided or have the dosage adjusted?

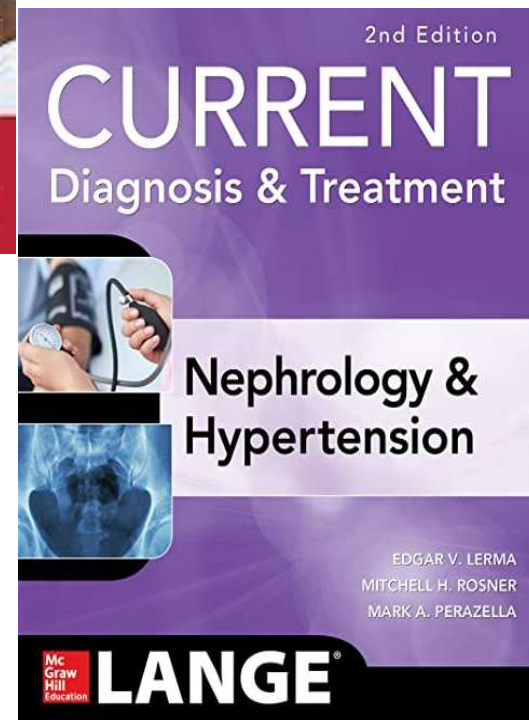
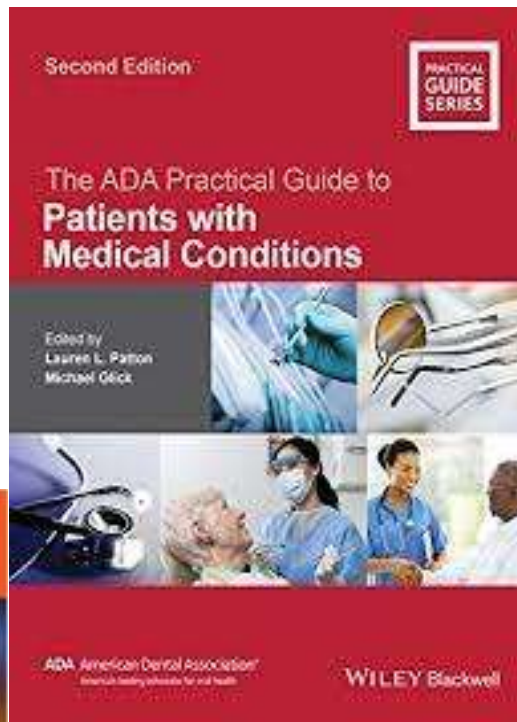
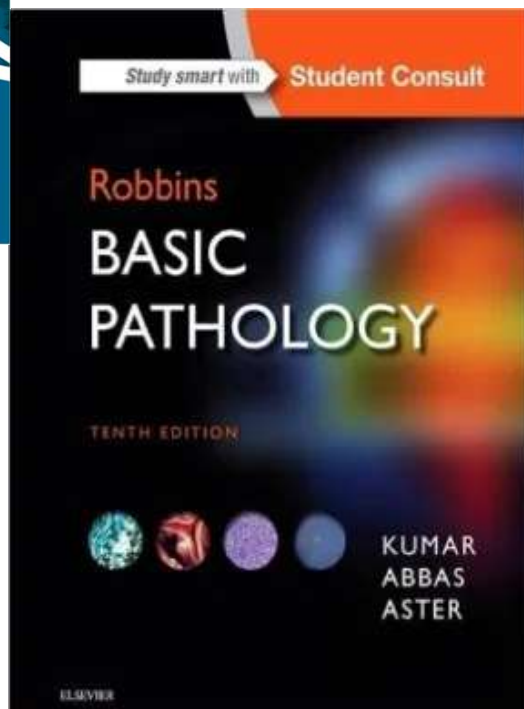
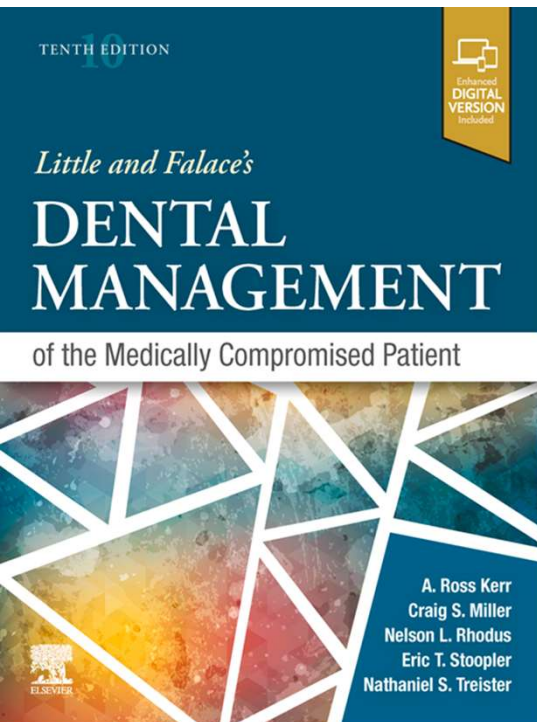
Dental Modifications for Chronic Kidney Disease

- Complication most likely to arise in patients in the late stages of CKD
- If the GFR is below 50 mL/min
 - Medication dosage may require adjustments
 - Avoid nephrotoxic medications when possible
 - High dosages of acetaminophen, NSAIDs, aminoglycoside antibiotics, tetracycline and acyclovir (anti-viral)
- For patients receiving dialysis
 - Invasive dental treatment should NOT occur on the dialysis treatment days
 - Avoid taking blood pressure on the arm where the AV shunt is placed
- Kidney transplant
 - Use of immunosuppressant therapy after transplant placement
 - Risk acquiring infections

Drug and Usual Dose	Dosage Adjustment for Renal Failure ^a		Removed by Dialysis
	GFR (mL/min)		
	10–50	<10	
Analgesics			
Acetaminophen 650mg q4h	No adjustment needed	q8h	HD: Yes PD: No
Aspirin 650mg q6h	50%	Avoid	Yes ^b
Celecoxib (Celebrex) 100–200mg q12h	Avoid if GFR <30mL/min	Avoid	No
Codeine 30–60mg q4–6h	75%	Avoid	No
Ibuprofen (Motrin) 400–800mg q8h	No adjustment needed	Avoid	No
Meperidined (Demerol) 50mg q4h	75%	50%	No
Tramadol (Ultram) 50–100mg q6h	q6–12h	50% q12h	No ^a
Anesthetics			
Articaine, lidocaine, mepivacaine, prilocaine	No adjustment needed	No	ND
Adjunctive analgesic			
Gabapentin (Neurontin) 200–600mg q8h	200–600mg q12–24h	<100mg once daily	Yes ^b
Antimicrobials			
Acyclovir–(Zovirax) 200–800mg q4h	q8h	q12h	Yes ^b
Amoxicillin 500mg q8h	q8–12h	q12–24h	Yes ^a
Azithromycin (Zithromax) 250–500mg q24h	No adjustment needed	Avoid	ND
Cephalexin (Keflex) 250–500mg q6h	q6–8h	q12–24h	Yes ^b
Clarithromycin 250mg q12h	50%–100% q12h	50% q12h	ND
Clindamycin (Cleocin) 150–300mg q6h	No adjustment needed	No	ND
Doxycycline (Vibramycin) 100mg q12h	No adjustment needed	No	ND
Erythromycin 250–500mg q6h	No adjustment needed	No	ND
Fluconazole (Diflucan) 100–200mg q24h	50%	25%	Yes ^b
Metronidazole (Flagyl) 250–500mg q8–12h	No adjustment needed	Yes ^a	ND
Tetracycline ^c (Sumycin, Aureomycin) 250–500mg q6–12h	Avoid	Avoid	No
Benzodiazepine			
Diazepam (Valium), ^d 2–5mg q12h, triazolam (Halcion) 0.125mg at bedtime	No adjustment needed	No	ND
Corticosteroid			
Dexamethasone, hydrocortisone, prednisone 5–10mg/day	No adjustment needed	No	ND
Sedative Hypnotic			
Chloral hydrate 250–500mg/day	Contraindicated	Yes	ND

Drug Dosage Adjustment for Renal Failure

Dental Management of the Medically Compromised Patient 10th Ed.



Questions?